

5.2.64

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Question

Solve for the following system of linear equations:

$$\mathbf{X} + \mathbf{Y} = \begin{pmatrix} 7 & 0 \\ 2 & 5 \end{pmatrix}$$

$$\mathbf{X} - \mathbf{Y} = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$$

Solution

The given system can be represented as:

$$\begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 2 & 5 \\ 3 & 0 \\ 0 & 4 \end{pmatrix} \quad (1)$$

Solving the equation,

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}^{-1} \begin{pmatrix} 7 & 0 \\ 2 & 5 \\ 3 & 0 \\ 0 & 4 \end{pmatrix} \quad (2)$$

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 7 & 0 \\ 2 & 5 \\ 3 & 0 \\ 0 & 4 \end{pmatrix} \quad (3)$$

Solution

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 1 & 9/2 \\ 2 & 0 \\ 1 & 1/2 \end{pmatrix} \quad (4)$$

Hence, the matrices are given by

$$\mathbf{X} = \begin{pmatrix} 5 & 0 \\ 1 & 9/2 \end{pmatrix} \quad (5)$$

$$\mathbf{Y} = \begin{pmatrix} 2 & 0 \\ 1 & 1/2 \end{pmatrix} \quad (6)$$

Results

Matrix	Value
X	$\begin{pmatrix} 5 & 0 \\ 1 & 9/2 \end{pmatrix}$
Y	$\begin{pmatrix} 2 & 0 \\ 1 & 1/2 \end{pmatrix}$

Table: Results